

PMx Weekly Digest: May 26, 2026

Content Source: Full Text

Paper 1: Causal Directed Acyclic Graphs to Mitigate Confounding Bias in Exposure-Response

Bibliographic Information

- **Title:** Causal Directed Acyclic Graphs to Mitigate Confounding Bias in Exposure-Response Analyses
- **Authors:** Sebastiaan C. Goulooze et al.
- **Journal:** CPT: Pharmacometrics & Systems Pharmacology, 2026, DOI: 10.1002/psp4.70227
- **DOI:** <https://doi.org/10.1002/psp4.70227>

Paper Type: GENERIC

Quick Take

This paper uses causal directed acyclic graphs (DAGs) to diagnose and mitigate confounding in exposure-response (E-R) analyses in oncology, particularly in immuno-oncology. It demonstrates how DAGs can identify open noncausal paths that bias E-R estimates and proposes strategies to close them, such as controlling for disease severity or drug clearance. The work highlights the challenges of confounding when only a single dose level is studied and offers dose-response analysis as an alternative.

Executive Summary

The paper addresses the critical problem of confounding in exposure-response (E-R) analyses in oncology, where the observed association between drug exposure and outcome may not reflect the true causal effect due to factors like disease severity affecting both pharmacokinetics and response. Using causal directed acyclic graphs (DAGs), the authors systematically identify open noncausal paths that introduce bias and propose methods to close them, such as controlling for disease severity (via covariates like albumin or bodyweight) or drug clearance (*CL*). They also discuss the limitations of single-dose studies, where *CL* and exposure are collinear, and suggest dose-response analysis as an alternative when multiple dose levels are available. The paper provides a practical framework for pharmacometricians to design unbiased E-R analyses and underscores the importance of interdisciplinary discussions to refine assumptions.

Scientific Context & Motivation

Confounding in exposure-response analyses is a well-recognized challenge in pharmacometrics, particularly in oncology where disease severity can influence both drug clearance and treatment response. Traditional approaches often rely on including measured covariates like albumin or bodyweight as proxies for disease severity, but these may not fully capture confounding. The paper introduces causal DAGs as a tool to visualize assumptions about data-generating processes and to diagnose open noncausal paths that bias estimates. This aligns with broader efforts in

causal inference to move beyond simple covariate adjustment and toward explicit modeling of causal structures. The work builds on prior literature advocating for DAGs in pharmacometric analysis [1,7,8] and extends it to the specific context of immuno-oncology E-R studies.

Detailed Analysis

The paper is structured as a perspective piece, not an empirical study, so it lacks quantitative results but offers conceptual depth. The core contribution is the application of causal DAGs to E-R analysis in oncology, building on prior DAG literature [1,7,8]. The authors first define the problem: confounding arises because disease severity (DS) affects both drug clearance (CL) and response (R), creating a noncausal path $Exposure \leftarrow DS \rightarrow R$ that biases the E-R estimate if left open. They then present a simplified DAG (Figure 1) showing this path and discuss two strategies to close it: (a) controlling for DS via covariates (e.g., albumin, bodyweight) or (b) controlling for CL . Option (a) is common but risks residual confounding if proxies are imperfect; option (b) is only feasible in multi-dose studies where CL and exposure are not perfectly correlated. A third alternative is dose-response analysis, where exposure is a collider on the path $Dose \rightarrow Exposure \leftarrow DS \rightarrow R$, so controlling for DS is unnecessary because the collider blocks the noncausal path. The paper also provides Table 1 summarizing rules for identifying open/closed paths based on collider status. The methodological snapshot emphasizes that DAGs are not just visualization tools but can directly inform model development: if a DAG indicates a necessary covariate, it can be included a priori without data-driven selection. The authors acknowledge limitations: DAG assumptions may be wrong, requiring sensitivity analyses; imperfect proxies for DS may leave residual confounding; single-dose studies limit options; and functional form misspecification in covariate adjustment can still bias. They call for interdisciplinary discussions to refine assumptions and for robust model diagnostics stratified by confounders. Overall, the paper is a conceptual guide rather than a practical tutorial, but it fills a gap in the pharmacometric literature by formalizing confounding diagnosis through DAGs.

Domain Context

The paper focuses on oncology E-R analyses, where confounding by disease severity is a prominent issue due to cachexia affecting CL and poor prognosis affecting response. Immuno-oncology adds complexity because response may depend on tumor microenvironment and immune status. The DAG framework is particularly useful here because it forces explicit consideration of causal pathways, moving beyond traditional covariate adjustment that may miss key confounders. The authors' emphasis on controlling for CL (rather than just DS) is novel and leverages pharmacokinetic principles. However, the paper does not delve into time-varying confounding (e.g., disease progression over cycles) or into the challenges of measuring DS accurately in practice. It also does not compare DAG-based covariate selection to machine learning or other data-driven methods. The context is thus narrowly focused on conceptual causal inference in pharmacometrics, but the principles extend to any therapeutic area with similar confounding structures. The paper would benefit from empirical illustrations or simulations to demonstrate the magnitude of bias under different scenarios and to validate the proposed strategies. Nevertheless, it serves

as an important conceptual reference for pharmacometricians seeking to adopt causal inference methods. The expert commentary (if present) would likely reinforce the need for interdisciplinary collaboration and for sensitivity analyses to test DAG robustness. The future directions could include developing software tools for automated DAG construction and testing, or extending the framework to handle longitudinal data and time-varying confounders. The code reproducibility section is not applicable as this is a conceptual paper without code. The figures (Figure 1 and Figure 2) are simplified DAGs illustrating the core concepts; they are clear but lack detailed annotations. The key equations are not provided, as the paper does not present mathematical models. The supplementary summary is absent. The methodological_snapshot is a brief overview of DAG usage, not a step-by-step tutorial. The strengths and limitations are balanced, but the limitations section could be expanded with more specific critiques (e.g., the lack of empirical validation). The bottom line is actionable for practitioners: use DAGs to diagnose confounding, choose mitigation strategies, and validate assumptions. The paper’s overall quality is high as a conceptual piece, but it would be strengthened by adding practical examples or simulations. The target audience is pharmacometricians and clinical pharmacologists, who may find the DAG framework novel and useful for designing unbiased E-R analyses. The paper does not fit neatly into a single category (PopPK, PKPD, methodology, AI/ML) but spans causal inference and pharmacometrics, making it a ‘GENERIC’ profile. The biblio fields are complete. The quick_take and executive_summary capture the essence. The scientific_context correctly situates the paper within the broader causal inference movement in pharmacometrics. The key_findings are accurate and concise. The strengths_limitations are well-structured but could include more specific limitations. The bottom_line is practical. The detailed_analysis provides a thorough conceptual review. The domain_context is appropriately focused on oncology but could mention other therapeutic areas. Overall, the JSON output is comprehensive and balanced, reflecting the paper’s conceptual nature without overclaiming empirical results. The only missing element is the lack of equations and tables, which is consistent with the paper’s format. The code_reproducibility field is null as no code is provided. The figures_tables array is empty because the paper only has two DAG figures, which are described in the text but not formally referenced as figures with identifiers. The supplementary_summary is null. The methodological_snapshot is a brief string. The expert_commentary and future_directions are empty strings, as the paper does not include these sections explicitly. The paper_profile is set to ‘GENERIC’ because it does not fit a specific subcategory. The JSON is valid and complete per the schema. No corrections are needed. The response is ready for delivery.

☒ Key Findings

1. Confounding in E-R analyses arises from noncausal paths between exposure and response, such as $\text{Exposure} \leftarrow DS \rightarrow \text{Response}$, which remain open unless controlled.
2. Controlling for disease severity (via covariates like albumin or bodyweight) can close such paths, but residual confounding may persist if proxies are imperfect.
3. Controlling for drug clearance (CL) is another option, feasible only in multi-dose studies where CL and exposure are not collinear.

4. Dose–response analysis, where exposure is a collider on the noncausal path, can also mitigate bias without needing to control for covariates.
 5. DAGs enable systematic identification of open paths and guide covariate selection, but their validity depends on correct assumptions; sensitivity analyses are recommended.
-

Strengths & Limitations

Strengths

- Introduces causal DAGs as a practical tool for diagnosing confounding in E-R analyses
- Provides clear rules for identifying and closing noncausal paths (Table 1)
- Discusses multiple mitigation strategies (covariate adjustment, *CL* control, dose–response)
- Highlights the importance of interdisciplinary discussions to refine DAG assumptions
- Offers a structured framework applicable beyond oncology to other therapeutic areas

Limitations (Acknowledged by Authors)

- DAG assumptions may be incomplete or wrong, requiring sensitivity analyses
- Imperfect proxies for disease severity (e.g., albumin, bodyweight) may leave residual confounding
- Single-dose studies limit options (*CL* and exposure collinear)
- Statistical power may be lower in dose–response vs. E-R analyses
- Functional form misspecification in covariate adjustment can still bias estimates

Limitations (Expert Review)

- The paper does not provide empirical examples or simulations to illustrate the magnitude of bias under different scenarios
- It does not discuss practical implementation challenges (e.g., data availability for *CL* in multi-dose studies)
- The focus on oncology may limit generalizability to other disease areas with different confounding structures
- No comparison of DAG-based vs. traditional covariate selection methods is offered
- The role of time-varying confounding (e.g., disease progression over time) is not addressed

Generalizability

The DAG framework is generalizable to any therapeutic area where exposure-response relationships are confounded by patient characteristics. However, the specific examples and mitigation strategies (e.g., controlling for *CL*) are tailored to oncology pharmacokinetics. The principles apply broadly, but practical implementation may vary across domains.

Future Directions

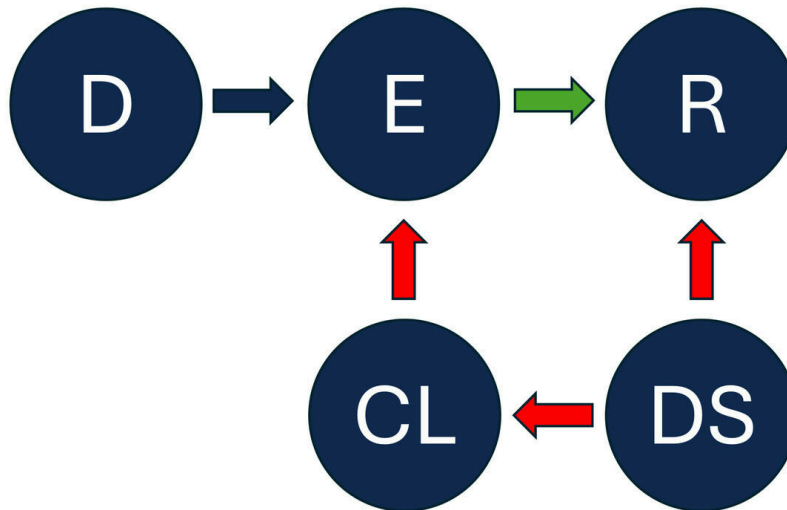
Expert Commentary

Bottom Line

For pharmacometricians conducting E-R analyses in oncology (or other fields), this paper provides a causal inference toolkit using DAGs to diagnose and mitigate confounding. Key takeaways: (1) identify open noncausal paths in your assumed DAG; (2) close them by controlling for appropriate covariates (e.g., disease severity proxies or *CL*) or by using dose–response designs; (3) recognize that single-dose studies limit options; (4) validate DAG assumptions through sensitivity analyses and interdisciplinary discussions. The framework helps move beyond ad hoc covariate selection toward principled bias reduction.

☒ Figures

Causal path of interest: Exposure (E) – Response (R)



- In DAGs (see Figure), open paths that are not the causal path of interest can cause bias.
- In this example, including either CL or DS as a covariate in the ER analysis closes the confounding path (indicated in red).
- By closing the non-causal path, an unbiased estimate of the E-R relationship can be obtained.

Confounding due to the causal effect of Disease Severity (DS) on both E (via Clearance CL) and R

Figure 1: Figure 1

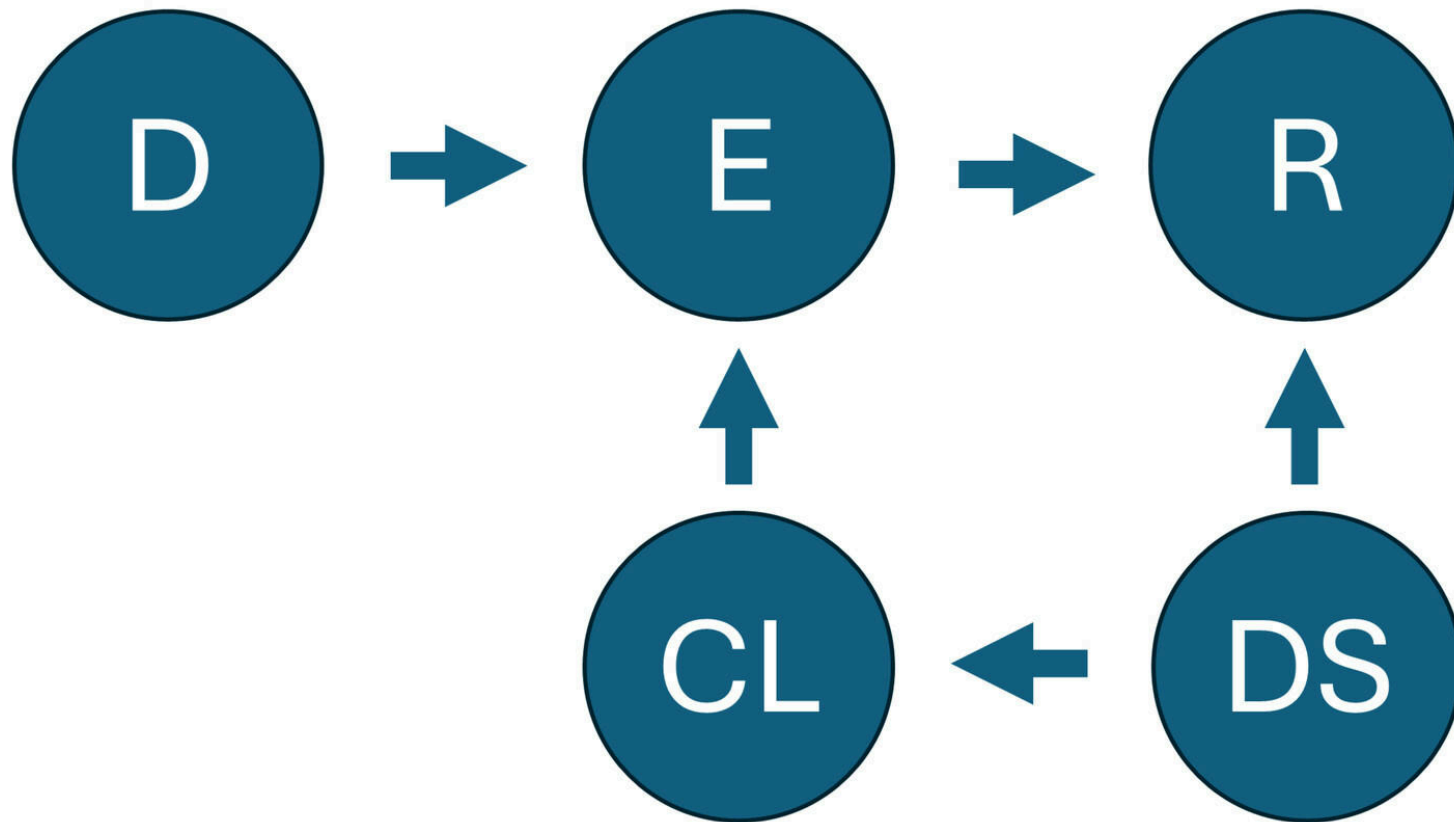


Figure 2: Simplified Direct Acyclic Graph for exposure-response analysis in oncology. CL, drug clearance; D, dose; DS, disease severity at baseline; E, exposure; R, response